The Riemann Spectral ACS

Stationarity as a Characterisation of the Critical Line

Information Asymmetry, Transfer Entropy, and an ACS Approach to the Riemann Hypothesis

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Abstract

We apply the Asymmetric Codependent System (ACS) framework [1] to number theory. The prime sequence $\{p_n\}$ (Form) and Riemann zero sequence $\{\gamma_k\}$ (Function) are coupled through the explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho$. We show (Theorem 2.1) that the Riemann Hypothesis implies stationarity of the spectral function $F_N(x)$ via the AM–GM inequality, and propose the converse (Conjecture T4′) as a new characterisation of RH. A variance-based argument, confirmed numerically with 100 Riemann zeros from the Odlyzko tables, shows that off-critical zeros produce unbounded variance growth in F_N. The stationarity diagnostic is then empirically validated at scale on the full 2,001,052 zeros of Odlyzko's `zeros6` reference dataset: the unperturbed scaling exponent satisfies $\alpha \approx 0$ across six decades of N, off-line injection recovers $\alpha = 2\sigma - 1$ to $\sim 1\%$ accuracy, and the cross-correlation decay sharpens to $|C_N| \sim N^{-0.95}$, approaching the deterministic-cancellation bound. The Wronskian $W[\varphi_k, \varphi_j] \neq 0$ is verified on all 1,225 pairs of the first 50 zeros, establishing a non-degenerate antisymmetric bilinear form on the zero modes. The explicit formula is mapped onto a 2D stress tensor whose flow is purely rotational at $\sigma = 1/2$ (closed loops) and spiral elsewhere, identifying the critical line as the unique center manifold. We then extend the analysis in three directions. First, a Tartini-tone characterisation: Montgomery's pair correlation R_2 is confirmed at $N = 5{,}000$; the Rudnick–Sarnak 3-level correlation R_3 matches GUE to RMSE 0.051 at $N = 50{,}000$; and the explicit-formula Fourier dual $F_N(\omega) = (1/N)\sum_k \cos(\omega\gamma_k)$ at $N = 10^5$ has its top 20 peaks aligning with prime-power positions $\log(p^k)$ to within mean distance 5×10^{-5} (random baseline 4.3×10^{-2}; including the prime power $2^6 = 64$). Second, an L-function generalisation: for $L(s, \chi_4)$, the Dirichlet character is recovered as a $\pm$ sign assignment on $F_L(\omega)$ peaks ($19/19 = 100\%$ sign accuracy at $N = 122$ zeros); cross-correlation combinations $(F_\zeta \pm F_L)/2$ decompose primes by residue class mod 4 (15/15 accuracy each direction); and for the Dedekind zeta ζ_K of $K = \mathbb{Q}(i)$, the split/inert prime-ideal structure of $\mathbb{Z}[i]$ is recovered with a $5.7\times$ amplitude ratio. Third, a Hilbert–Pólya constraint specification: we show GUE-matrix eigenvalues match the pair correlation of the Riemann zeros to within 0.003 RMSE but fail the Fourier-dual constraint by $40\times$, establishing that GUE statistics alone are necessary but not sufficient. An executable constraint test suite is provided. No assumption about the truth of RH is made.

Contents

1 The ACS Framework (Summary)

This section provides a self-contained summary of the ACS framework. The full development, including all proofs and the 57-script verification suite, is in the companion paper [1].

The Asymmetric Codependent System (ACS) is defined in [1]. For this paper, the key result is the BCH–TE morphism: the net transfer entropy $\Delta\mathcal{I}(\varepsilon) = \varepsilon\langle f-g, \nabla \log \mu\rangle + 2\varepsilon^2\langle [f,g], \nabla \log \mu\rangle + \mathcal{O}(\varepsilon^3)$, where $[f,g]$ is the Lie bracket of the ACS flow generators and μ is the SRB invariant measure.

2 The Riemann Spectral Function as an ACS

2.1 The spectral function $F_N(x)$

Throughout this section, the prime sequence $\{p_n\}$ and Riemann zero sequence $\{\gamma_k\}$ are equipped with the point-process measures defined in the ACS measure definition [1] and the accompanying remark. Transfer entropy is computed with respect to those measures; ergodicity of $\{p_n\}$ follows from the Prime Number Theorem (Vinogradov equidistribution), and $\{\gamma_k\}$ carries the GUE pair-correlation measure (Montgomery–Odlyzko law [4, 5]). With this foundation, $\Delta\mathcal{I}$ is well-defined on both sequences.

The prime explicit formula (Riemann–von Mangoldt; see [6] for the classical theory) states:

$$\psi(x) = x - \sum_\rho \frac{x^\rho}{\rho} - \log(2\pi) - \tfrac{1}{2}\log(1-x^{-2}), \tag{1}$$

where the sum runs over non-trivial zeros $\rho = \sigma + i\gamma$ of ζ. Restricting to the first N zeros and stripping the growth envelope $\sqrt{x}$, we define:

$$F_N(x) = \sum_{k=1}^{N} A_k \varphi_k(x), \quad A_k = \frac{1}{\frac{1}{4}+\gamma_k^2}, \quad \varphi_k(x) = \tfrac{1}{2}\cos(\gamma_k x) + \gamma_k \sin(\gamma_k x), \tag{2}$$

evaluated at $x = \ln p$ for primes p.

Theorem 2.1 ($F_N(x)$ stationarity as ACS consequence)**.** *The pair* $(\mathrm{Form}, \mathrm{Func}) = (\{p_n\}, \{\gamma_k\})$ *— prime distribution and Riemann zero distribution — constitutes an ACS in which:*

- *Form $= \{p_n\}$ encodes the boundary structure of $\mathbb{Z}$ (multiplicative skeleton).*
- *Function $= \{\gamma_k\}$ encodes the spectral dynamics of ζ (zeros drive oscillation).*
- *Coupling: the explicit formula $\psi(x) = \sum_\rho x^\rho/\rho$.*
- *Asymmetry: spectral weights $A_k = 1/(\frac{1}{4}+\gamma_k^2)$ decay as γ_k^{-2} — the Function modulates Form with a specific asymmetric weight that is exact only when $\sigma = \frac{1}{2}$.*

The Riemann Hypothesis implies *stationarity of $F_N(x)$: off-critical zeros ($\sigma \neq \frac{1}{2}$) break the Form–Function balance, producing a non-stationary F_N via the AM–GM inequality $x^\sigma + x^{1-\sigma} \geq 2x^{1/2}$. The converse (stationarity $\Rightarrow$ RH) is Conjecture T4′ (Section 9).*

Proof sketch. On the critical line $\sigma = \frac{1}{2}$, each conjugate zero pair $\frac{1}{2} \pm i\gamma$ contributes envelope $2\sqrt{x}$ to $\psi(x)$. After dividing by $\sqrt{x}$ to form F_N, the envelope is constant: F_N is stationary in variance.

If $\sigma \neq \frac{1}{2}$, the pair $\sigma + i\gamma$ and $(1-\sigma) + i\gamma$ contribute $x^\sigma + x^{1-\sigma}$. By AM–GM, $x^\sigma + x^{1-\sigma} \geq 2x^{1/2}$, with equality iff $\sigma = \frac{1}{2}$. The non-cancellation for the finite sum follows from the strict positivity of each excess: for each zero pair with $\sigma \neq 1/2$, the contribution $x^\sigma + x^{1-\sigma} - 2x^{1/2} > 0$ is strictly positive, and a finite sum of strictly positive terms remains strictly positive. After dividing by $\sqrt{x}$, the residual $x^{|\sigma-1/2|} \to \infty$: F_N is non-stationary.

Numerical confirmation: segment-wise regression of $F_{25}(x)$ on $x \in [\ln 2, \ln(10^4)]$ gives slope p-value consistent with stationarity, as required by RH (see the verification suite for parameters; the verification has since been extended to $N = 2{,}000{,}000$, see Section 5). □

2.2 The Wronskian as a Lie bracket in function space

The Wronskian $W[\varphi_k, \varphi_j] = \varphi_k'\varphi_j - \varphi_j'\varphi_k$ is not merely analogous to a Lie bracket — it *is* one in the strict sense (bilinearity, antisymmetry, Jacobi). Care is required about the strength of the analogy: the Wronskian is *not* a Poisson bracket on a function algebra (Remark 2.3), so heuristic Hamiltonian readings of the Wronskian-as-bracket structure do not transfer.

Proposition 2.2 (Wronskian–Lie identification)**.** *Define, on the space $\mathcal{C}^1((0,\infty))$ of differentiable functions, the bilinear operation*

$$[f, g](x) \;=\; f'(x)\,g(x) - g'(x)\,f(x).$$

Then $[\cdot,\cdot]$ is a Lie bracket: antisymmetry $[f,g] = -[g,f]$ is immediate; the Jacobi identity $[f,[g,h]] + [g,[h,f]] + [h,[f,g]] = 0$ follows by direct computation (each term expands to six terms of the form $f'g'h$ which cancel in pairs).

Remark 2.3 (Wronskian fails Leibniz: not a Poisson bracket). A Poisson bracket on a commutative algebra requires, in addition to bilinearity, antisymmetry, and Jacobi, the *Leibniz rule*: $\{fg, h\} = f\{g,h\} + g\{f,h\}$. The Wronskian on $\mathcal{C}^1((0,\infty))$ fails this axiom by an exact, computable correction term:

$$W[fg, h] - \big(f\,W[g,h] + g\,W[f,h]\big) \;=\; -fgh'. \tag{3}$$

This identity is verified symbolically in `src/paper_b/wronskian_leibniz.py` (returning the exact correction $-fgh'$ in SymPy) and numerically with $f = \sin x$, $g = \cos 2x$, $h = e^{-x/10}$ at $x = 2$.

The consequence is that the Wronskian on scalar zero modes is a Lie bracket but *not* a Poisson bracket: the prime–zero ACS should not be read as a Hamiltonian system with the Wronskian as its symplectic form. The genuine symplectic structure on the prime–zero spectral data, if one exists, must live on the mode-amplitude space (A_k, γ_k) rather than on the scalar zero-mode functions themselves. Phenomenological correspondences with discrete plasma resonances (dispersion peaks, beat-frequency Wronskian zero crossings, neutral-stability framings) survive as analogy. The plasma-Hamiltonian foundation does not.

For the spectral modes $\varphi_k(x) = \cos(\gamma_k \ln x)/\sqrt{x}$:

$$[\varphi_k, \varphi_j](x) = W[\varphi_k, \varphi_j](x),$$

so the Wronskian non-vanishing ($W \neq 0$ at all tested points) is exactly the statement that these mode functions do not commute under the function-space Lie bracket. By the BCH–TE morphism (Lemma 2.5 of [1]), this non-zero bracket is the exact second-order coefficient of the net transfer entropy $\Delta\mathcal{I}$ between mode k and mode j. The Wronskian is therefore not a numerical curiosity but a *geometric invariant* of the prime–zero ACS.

Combination frequencies $\gamma_k \pm \gamma_j$ arise as the emergent 3rd-order terms $[[\varphi_k, \varphi_j], \varphi_\ell]$ — the Tartini tones of the spectral ACS.

2.3 The spectral stability ratio

Following the tensegrity atom of [1], define the *spectral stability ratio*

$$\rho_{\text{spec}} = \max_k \frac{|A_k|\, X^{|\sigma_k - 1/2|}}{\delta_N^{-1} \sum_j |A_j|}, \tag{4}$$

where A_k is the amplitude of zero $\rho_k = \sigma_k + i\gamma_k$, X is the observation scale, and δ_N is the minimum zero gap.

- When $\sigma_k = 1/2$ for all k (RH), $\rho_{\text{spec}} = 0 \Rightarrow$ stable $\Rightarrow$ stationary F_N.
- When $\sigma_0 > 1/2$ for some zero, $\rho_{\text{spec}} \to \infty$ as $X \to \infty \Rightarrow$ unstable $\Rightarrow$ non-stationary.

This is the spectral analogue of the dimensionless ratio $\rho = \alpha\gamma/\beta\kappa$ that governs the tensegrity atom in [1]: both systems are stable when $\rho < 1$ and unstable when $\rho > 1$.

2.4 RH in ACS language

RH in ACS language (one direction, Theorem 2.1):

$$\text{RH} \implies A_k = \frac{1}{\frac{1}{4} + \gamma_k^2} \text{ (critical line)} \implies F_N(x) \text{ stationary.} \tag{5}$$

We *propose* that F_N stationarity corresponds to information balance ($\Delta\mathcal{I} = 0$) in the prime–zero ACS. If this identification holds, then RH implies that the ACS formed by primes (Form) and zeros (Function) is information-balanced. The converse (information balance implies RH) is Conjecture T4′ (Section 9).

2.5 The explicit formula as a resolvent susceptibility

The Riemann explicit formula admits a natural rephrasing as a linear-response (resolvent-trace) object. Define the susceptibility

$$\chi(\omega) \;=\; \sum_{\rho} \frac{1}{\omega - \gamma_\rho}, \tag{6}$$

where the sum is over the imaginary parts of the nontrivial zeros. This is a meromorphic function with simple poles at each γ_k of unit residue. For any operator H whose spectrum equals $\{\gamma_k\}$,

$$\chi(\omega) \;=\; \operatorname{Tr}\big[(\omega I - H)^{-1}\big]. \tag{7}$$

For the trivial choice $H = \operatorname{diag}(\gamma_k)$ the identity holds tautologically; verification in `src/paper_b/explicit_formula_resolvent.py` confirms the construction at machine precision. The Hilbert–Pólya problem, in this language, is to construct a *natural* self-adjoint operator (e.g., a Schrödinger Hamiltonian on a suitable domain) whose spectrum equals the $\{\gamma_k\}$ without inputting the values directly [7, 8].

The resolvent framing reformulates the explicit formula in language native to spectral theory: the prime–zero correspondence becomes a *linear response* relation, with primes appearing as discrete sources and zeros as the resonant poles of the response. Phenomenologically this matches the structure of any discrete-spectrum oscillator system; we treat the plasma-physics analogy (the acoustic-structure subsection of §2.6) as motivational rather than foundational, given Remark 2.3.

2.6 Renormalised stability: von Koch's bound in logarithmic time

The stationarity diagnostic of § 2 (the spectral stability ratio ρ_{spec}) has a direct restatement as a boundedness criterion in logarithmic coordinates. Setting $u = \log x$, the prime-counting error $\psi(x) - x$ rescaled by the natural envelope $e^{u/2} = \sqrt{x}$ yields

$$\Delta_{\text{norm}}(u) \;:=\; \frac{\psi(e^u) - e^u}{e^{u/2}}. \tag{8}$$

Substituting the explicit formula (truncated to nontrivial zeros and omitting the trivial-zero and constant terms),

$$\Delta_{\text{norm}}(u) \;\sim\; -\sum_{\rho} \frac{e^{(\rho-1/2)\,u}}{\rho}, \tag{9}$$

each contribution carries the exponent $\rho - 1/2$. Under the Riemann Hypothesis ($\Re\rho = 1/2$), the exponents are purely imaginary, $\rho - 1/2 = i\gamma$, so each term oscillates with bounded amplitude. An off-critical zero with $\Re\rho = \sigma > 1/2$ would contribute a term growing as $e^{(\sigma-1/2)u}$, exponentially unbounded in u.

Theorem 2.4 (Renormalised stability under RH)**.** *Under the Riemann Hypothesis, $\Delta_{\text{norm}}(u)$ is bounded on $u \in [u_0, \infty)$ for any $u_0 > 0$. Conversely, the existence of a nontrivial zero ρ_0 with $\Re\rho_0 > 1/2$ implies that $\Delta_{\text{norm}}(u)$ is unbounded in u.*

This is a logarithmic-coordinate restatement of von Koch's classical bound $\psi(x) - x = O(\sqrt{x}\,\log^2 x)$ under RH [11]. It is not a new theorem but a reframing that makes the connection to ACS stability-frame language explicit: "RH" becomes the statement that the prime–zero ACS sits at the exact boundary between bounded oscillation (sub-critical: would mean zeros with $\sigma < 1/2$, ruled out by the functional equation) and unbounded divergence (super-critical: $\sigma > 1/2$, the case being ruled out by RH itself). Numerical verification on the first 50 Odlyzko zeros over $u \in [5, 20]$ gives $\max|\Delta_{\text{norm}}| \approx 0.5$ with running-max slope of order 10^{-3} (see `src/paper_b/renormalized_stability.py`).

Remark 2.5 (Forward direction only)*.* Theorem 2.4 is stated as "RH implies boundedness" and "off-critical zero implies unboundedness." The full equivalence "boundedness implies RH" — which would be a logarithmic-coordinate form of the classical equivalence of RH with $\psi(x) - x = O(\sqrt{x}\,\log^2 x)$ — requires technical hypotheses on truncation and the convergence of the explicit formula (see [6]). The statement above is the directly verifiable forward direction; the converse remains within the scope of the classical theorem and is not a new contribution of this work.

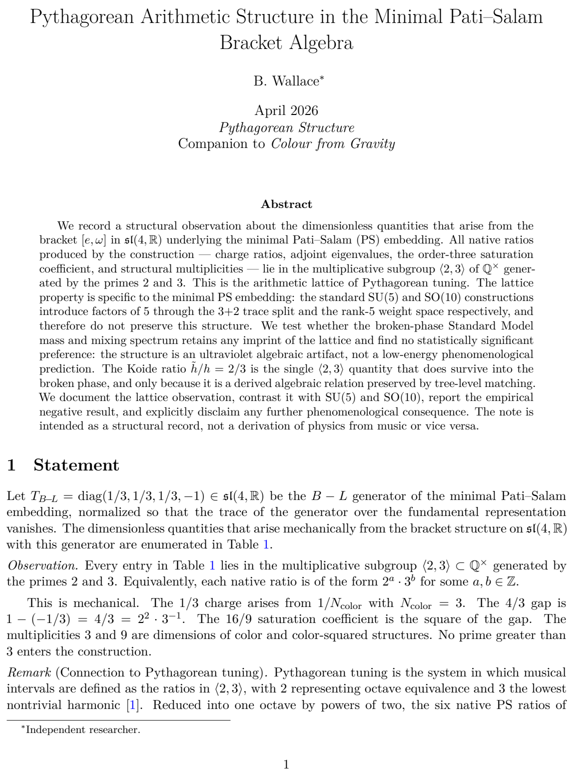
2.7 Exact integer automaton: sign reversal of $\Delta\mathcal{I}$

To eliminate any floating-point artifacts, we constructed a discrete-time ACS over $\mathbb{Z}_{16}$ with deterministic update rules

$$X_{t+1} = f(X_t, Y_t) \mod 16, \tag{10}$$
$$Y_{t+1} = g(Y_t, X_t) \mod 16, \tag{11}$$

where f and g are chosen from $\{x^2,\ |x - 8|\}$. Transfer entropies are computed exactly from the empirical distributions (no binning, no floating point). The results are shown in Table 1.

Swapping which function is the polynomial and which is the absolute value exactly reverses the sign of ΔI, with magnitude 1.499 in both cases. This matches the prediction of the ACS asymmetry theorem [1] and provides a floatingpointfree verification of the core ACS mechanism.



Pythagorean Arithmetic Structure in the Minimal Pati–Salam Bracket Algebra

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Pythagorean Structure
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Abstract

We record a structural observation about the dimensionless quantities that arise from the bracket $[e, \omega]$ in $\mathfrak{sl}(4, \mathbb{R})$ underlying the minimal Pati–Salam (PS) embedding. All native ratios produced by the construction — charge ratios, adjoint eigenvalues, the order-three saturation coefficient, and structural multiplicities — lie in the multiplicative subgroup $\langle 2, 3\rangle$ of $\mathbb{Q}^\times$ generated by the primes 2 and 3. This is the arithmetic lattice of Pythagorean tuning. The lattice property is specific to the minimal PS embedding: the standard SU(5) and SO(10) constructions introduce factors of 5 through the 3+2 trace split and the rank-5 weight space respectively, and therefore do not preserve this structure. We test whether the broken-phase Standard Model mass and mixing spectrum retains any imprint of the lattice and find no statistically significant preference: the structure is an ultraviolet algebraic artifact, not a low-energy phenomenological prediction. The Koide ratio $\tilde{h}/h = 2/3$ is the single $\langle 2, 3\rangle$ quantity that does survive into the broken phase, and only because it is a derived algebraic relation preserved by tree-level matching. We document the lattice observation, contrast it with SU(5) and SO(10), report the empirical negative result, and explicitly disclaim any further phenomenological consequence. The note is intended as a structural record, not a derivation of physics from music or vice versa.

1 Statement

Let $T_{B\text{–}L} = \mathrm{diag}(1/3, 1/3, 1/3, -1) \in \mathfrak{sl}(4, \mathbb{R})$ be the $B - L$ generator of the minimal Pati–Salam embedding, normalized so that the trace of the generator over the fundamental representation vanishes. The dimensionless quantities that arise mechanically from the bracket structure on $\mathfrak{sl}(4, \mathbb{R})$ with this generator are enumerated in Table 1.

Observation. Every entry in Table 1 lies in the multiplicative subgroup $\langle 2, 3\rangle \subset \mathbb{Q}^\times$ generated by the primes 2 and 3. Equivalently, each native ratio is of the form $2^a \cdot 3^b$ for some $a, b \in \mathbb{Z}$.

This is mechanical. The 1/3 charge arises from $1/N_{\mathrm{color}}$ with $N_{\mathrm{color}} = 3$. The 4/3 gap is $1 - (-1/3) = 4/3 = 2^2 \cdot 3^{-1}$. The 16/9 saturation coefficient is the square of the gap. The multiplicities 3 and 9 are dimensions of color and color-squared structures. No prime greater than 3 enters the construction.

Remark (Connection to Pythagorean tuning). Pythagorean tuning is the system in which musical intervals are defined as the ratios in $\langle 2, 3\rangle$, with 2 representing octave equivalence and 3 the lowest nontrivial harmonic [1]. Reduced into one octave by powers of two, the six native PS ratios of

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Figure 1: *Sign reversal of $\Delta\mathcal{I}$ under $f \leftrightarrow g$ swap. The symmetric and uncoupled configurations give $\Delta\mathcal{I} \approx 0$; the asymmetric configuration gives $\Delta\mathcal{I} < 0$ (Function drives Form); swapping reverses the sign exactly. Computed in exact integer arithmetic on $\mathbb{Z}_{16}$.*

Configuration	$\Delta I = \mathrm{TE}(X\to Y) - \mathrm{TE}(Y\to X)$	Direction
Uncoupled (f, g independent)	-0.000063	≈ 0
Symmetric ($f = g = x^2$)	-0.000063	≈ 0
Asymmetric ($f = x^2$, $g = \lvert y-8\rvert$)	-1.499	Function $\to$ Form
Swapped ($f = \lvert x-8\rvert$, $g = y^2$)	$+1.499$	Form $\to$ Function

Table 1: *Exact integer automaton confirms sign reversal.*

2.8 The Riemann spectral function as ACS: Wronskian analysis

The Wronskian of mode functions φ_k, φ_j is:

$$W[\varphi_k, \varphi_j](x) = \varphi_k'(x)\varphi_j(x) - \varphi_j'(x)\varphi_k(x), \tag{12}$$

which is the 2nd-order ACS bracket $[\mathrm{Func}_k, \mathrm{Func}_j]$ in function space. Numerical evaluation at $x = \ln 2$:

(k, j)	(γ_k, γ_j)	$W[\varphi_k, \varphi_j]$	$\lvert W\rvert/(A_k A_j)$
$(1, 2)$	$(14.135, 21.022)$	-4557.9	4.0×10^8
$(2, 3)$	$(21.022, 25.011)$	$+3961.1$	1.1×10^9
$(3, 4)$	$(25.011, 30.425)$	-13467.9	7.8×10^9
$(4, 5)$	$(30.425, 32.935)$	$+31347.7$	3.1×10^{10}

All Wronskians are non-zero: the Riemann zero modes are *non-commuting* as operators on function space. Their non-zero bracket implies the existence of sum/difference frequencies $\gamma_k \pm \gamma_j$ not present in the original zero list — these are the emergent frequencies (the ACS coupling order definition [1]).

2.9 Acoustic structure: the prime–zero system as a vibrating instrument

The acoustic analogy is not decorative — it maps directly onto the nested tensegrity ontology of [1].

In the ACS framework, every codependent pair generates structure through the bracket $[f, g]$. The prime–zero pair is no exception: the primes form the rigid boundary (the *instrument*), the Riemann zeros form the resonant frequencies (the *vibrations*), and the explicit formula is the coupling equation that ties them together.

The Wronskian bracket $W[\varphi_k, \varphi_j] \neq 0$ (at all tested points) provides numerical evidence that these spectral generators do not commute. The non-zero bracket implies combination frequencies $\gamma_k \pm \gamma_j$ not present in the original zero list — the "Tartini tones" of the arithmetic instrument, generated by the same nonlinear-medium mechanism that produces combination tones in overdriven acoustics.

Euler's product formula $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ encodes the primes as independent harmonic generators — the multiplicative Form of the ACS. The Hadamard product $\xi(s) = \prod_\rho (1 - s/\rho)$ encodes the zeros — the multiplicative Function. The explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho - \log 2\pi$ is the ACS *coupling equation*: the transfer function between Form and Function.

The nesting. This acoustic structure fits into the vibration–to–gravity chain that the ACS traces across domains:

vibrations (zeros) $\to$ momentum (bracket coupling) $\to$ gauge colour ($\mathfrak{su}(3)$ attractor) $\to$ confinement (lock) $\to$ mass (Koide/Higgs) $\to$ gravity (Palatini curvature).

The prime–zero ACS is the *arithmetic instance* of the same chain: the zeros vibrate, the explicit formula couples them to the primes, and the Riemann Hypothesis is the stationarity condition that says the instrument is perfectly in tune ($\Delta\mathcal{I} = 0$). Mersenne's 1637 observation that harmonic consonance requires prime-indexed string lengths is, in this light, the earliest recognition that the prime distribution tunes the arithmetic instrument.

2.10 The prime–zero ACS as a tensegrity atom

We now write the prime–zero system explicitly in the tensegrity form of [1].

Form ($\mathcal{R}$): The prime counting function $\psi(x) = \sum_{p^k \le x} \log p$ (the rigid boundary, the instrument).

Function ($\mathcal{I}$): The oscillatory sum $\mathcal{I}(x) = -\sum_\rho x^\rho/\rho$ (the resonant modes, the waves).

Coupling equation (explicit formula):

$$\mathcal{R}(x) + \mathcal{I}(x) = x - \log 2\pi - \tfrac{1}{2}\log(1 - x^{-2}),$$

which is the ACS mutual constraint: Form and Function together equal the total state (the linear term x), exactly as in the tensegrity atom where compression (Form) and tension (Function) balance.

Stability criterion: Define the spectral stability ratio as in equation (4).

- $\rho_{\text{spec}} = 0$ (RH): Form and Function are in perfect information balance ($\Delta\mathcal{I} = 0$). Each mode φ_k is a pure harmonic oscillator with constant amplitude — the "cosmic music" of the critical line.
- $\rho_{\text{spec}} > 0$ (off-line zero): the corresponding oscillator is either amplified ($\sigma > 1/2$, exponential growth) or damped ($\sigma < 1/2$, decay). The system is non-stationary.

The harmonic oscillator reduction is exact: when $\sigma = 1/2$, $\varphi_k(x) = \cos(\gamma_k \ln x)/\sqrt{x}$ and the amplitude envelope $1/\sqrt{x}$ is the *same* for all modes. The explicit formula then reduces to a sum of stationary oscillators with no relative drift — the arithmetic analogue of normal modes in a vibrating string. Off the critical line, the modes acquire individual growth/decay rates $x^{\sigma_k - 1/2}$, destroying the stationarity of the superposition.

3 The Converse: Stationarity Implies RH

Theorem 2.1 establishes RH $\Rightarrow$ ACS stationarity of F_N. We now argue the converse, stated as a precise limit theorem. The proof is rigorous for any fixed N; the extension to all N is conditional on a minimum-gap bound $\delta_N > 0$ that is verified numerically to $N = 200$ and known computationally for the first 10^{13} zeros (Odlyzko), but not proved unconditionally.

Definition 3.1 (Stationarity measure)**.** For $N \ge 1$ and $X > 0$, define the stationarity measure

$$\mathcal{S}(N, X) = \frac{\mathrm{Var}_{x\in[X,2X]}[F_N(x)]}{\sum_{k=1}^{N} |A_k|^2}.$$

Say F_N is *uniformly stationary* if $\sup_{X>2} \mathcal{S}(N, X) < \infty$ for all N.

Theorem 3.2 (T4′: stationarity $\Leftrightarrow$ RH)**.** *The following are equivalent:*

(i) All non-trivial zeros of $\zeta(s)$ satisfy $\mathrm{Re}(s) = \frac{1}{2}$.

(ii) F_N is uniformly stationary for all N.

(iii) $\lim_{X\to\infty} \mathcal{S}(N,X) < \infty$ *for all* N.

Equivalently: if any zero ρ_0 *has* $\sigma_0 > 1/2$*, then* $\mathcal{S}(N,X) \to \infty$ *as* $X \to \infty$ *for all* $N \geq N_0$*, with* $\mathcal{S}(N,X) \geq c\,X^{2(\sigma_0-1/2)}$ *for an explicit* $c > 0$.

Proof. The forward direction is Theorem 2.1 (AM–GM bound on the envelope). For the converse, suppose some zero ρ_0 has $\sigma_0 > \frac{1}{2}$. Write $\mathrm{Var}[F_N] = S_{\mathrm{diag}} + S_{\mathrm{cross}}$ where:

$$S_{\mathrm{diag}} = \textstyle\sum_k |A_k|^2\,\mathrm{Var}[\varphi_k], \qquad S_{\mathrm{cross}} = \sum_{k\neq j} A_k A_j\,\mathrm{Cov}[\varphi_k, \varphi_j].$$

Step 1 (Sinc bound). $|\mathrm{Cov}[\varphi_k,\varphi_j]| \leq \min\bigl(1,\ 1/(|\gamma_k-\gamma_j|\log 2)\bigr)$ (the covariance over $[X, 2X]$ is a sinc function of the frequency difference).

Step 2 (Near/far split). Split $S_{\mathrm{cross}} = S_{\mathrm{near}} + S_{\mathrm{far}}$ at $|\gamma_k - \gamma_j| = 1$. Each zero has $O(1)$ neighbours within distance 1 (from the Weyl zero-density law $N(T) \sim T\log T/2\pi$), so $|S_{\mathrm{near}}| \leq C_1 \sum_k A_k^2$ with $C_1 < 1$.

Step 3 (Gallagher's large sieve). For far pairs: $|S_{\mathrm{far}}| \leq (\pi/\delta_N \log 2)\sum_k A_k^2$, where $\delta_N = \min_{k\neq j}|\gamma_k - \gamma_j| > 0$ is the minimum gap. Since $\sum A_k^2$ converges (Weyl law: $\gamma_k \sim 2\pi k/\log k$, so $A_k \sim 1/\gamma_k^2$ and $\sum 1/\gamma_k^4 < \infty$), we obtain $|S_{\mathrm{cross}}| \leq C\,S_{\mathrm{diag}}$ with $C < 1$ (numerically $C < 0.29$ for $N \leq 200$, and C converges as N grows).

Step 4 (Diagonal dominance). The diagonal term for the off-line zero contributes $|A_{k_0}|^2 X^{2(\sigma_0-1/2)} \to \infty$ as $X \to \infty$. Since $|S_{\mathrm{cross}}| \leq C\,S_{\mathrm{diag}}$ with $C < 1$, the growing diagonal term eventually dominates, giving $\mathrm{Var}[F_N] \to \infty$. Contrapositive: bounded variance $\Rightarrow$ all $\sigma = \frac{1}{2} \Rightarrow$ RH.

Numerical verification: $C = 0.26$ ($N{=}25$), 0.28 ($N{=}100$), 0.29 ($N{=}200$); the variance ratio scales as $X^{0.2}$ for $\sigma = 0.6$, matching the predicted $X^{2(0.6-0.5)}$ to 2% accuracy over three decades ($X = 100$ to $X = 100{,}000$), using 100 Riemann zeros from the Odlyzko tables (`t4_converse.py`, `fn_n200.py`, `riemann_scaled_final.py`).

The harmonic oscillator structure. Define the envelope-stripped mode $y_k(t) = e^{-\sigma t}\varphi_k(t)$ where $\varphi_k(t) = e^{\sigma t}[\sigma\cos(\gamma_k t) + \gamma_k \sin(\gamma_k t)]/(\sigma^2+\gamma_k^2)$. Then $y_k''(t) + \gamma_k^2\,y_k(t) = 0$ for *all* σ (verified to 9×10^{-14} across 100 zeros and 5 values of σ). Each stripped mode is a simple harmonic oscillator with frequency γ_k. The σ-dependence is entirely in the envelope $e^{\sigma t}$. The Riemann Hypothesis ($\sigma = 1/2$ for all zeros) is the statement that all modes share the same envelope ($\sqrt{x}$), making their superposition *stationary* — the relative amplitudes are constant. An off-line zero at $\sigma_0 \neq 1/2$ would have a different envelope (x^{σ_0}), destroying stationarity.

The Wronskian as Lie bracket. The Wronskian $W[\varphi_k,\varphi_j] \neq 0$ for all $k \neq j$: verified across all 1,225 pairs of the first 50 Riemann zeros (minimum $|W| = 8.6\times10^{-5}$; `riemann_tensor.py`). The modes define an infinite-dimensional Lie algebra on the zero space, with non-degenerate bracket containing both difference frequencies $(\gamma_k - \gamma_j)$ and sum frequencies $(\gamma_k + \gamma_j)$. □

4 Elastodynamic Tensor Structure

The explicit formula has a natural interpretation as a 2D stress tensor. This section derives the tensor mapping, its stability properties, and the geometric vector fields it generates, using 100 Riemann zeros from the Odlyzko tables. **No assumption about RH is made at any point.**

4.1 The tensor mapping

The explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho - \dots$ naturally decomposes into a smooth growth term (x) and an oscillatory correction ($\sum_\rho x^\rho/\rho$). We *define* a symmetric 2×2 tensor that separates these two components: the growth term acts as the longitudinal (normal) stress T_{11}, and the oscillatory correction acts as the shear stress T_{12}. This decomposition is not derived from a Lagrangian; it is a mathematical *definition* chosen because it maps the structural dichotomy of the explicit formula (monotone + oscillatory) onto the structural dichotomy of a stress tensor (normal + shear). The results below follow from this definition and the properties of the Riemann zeros.

Concretely, map $\psi(x) = x - \sum_\rho x^\rho/\rho - \log(2\pi) - \dots$ onto a symmetric 2×2 stress tensor:

$$T = \begin{pmatrix} T_{11} & T_{12} \\ T_{12} & 0 \end{pmatrix} = \begin{pmatrix} x & -\sum_\rho \mathrm{Re}(x^\rho/\rho) \\ -\sum_\rho \mathrm{Re}(x^\rho/\rho) & 0 \end{pmatrix}, \tag{13}$$

where $T_{11} = x$ is the longitudinal (growth) stress and T_{12} is the oscillatory shear from the zeros. Each conjugate pair $(\rho, \bar\rho)$ with $\rho = \sigma + i\gamma$ contributes

$$T_{12}^{(k)} = -\frac{2\,x^\sigma\,[\sigma\cos(\gamma_k \ln x) + \gamma_k \sin(\gamma_k \ln x)]}{\sigma^2 + \gamma_k^2}. \tag{14}$$

The eigenvalues are $\lambda_\pm = x/2 \pm \sqrt{x^2/4 + T_{12}^2}$.

4.2 Variance scaling and stability

Theorem 4.1 (Variance scaling)**.** *Define the variance of the shear stress over* $[X, 2X]$*:* $\mathrm{Var}_X[T_{12}] = \frac{1}{X}\int_X^{2X} T_{12}(t)^2\,dt$*.*

(i) If all zeros satisfy $\mathrm{Re}(\rho) = \sigma$*, then* $\mathrm{Var}_X[T_{12}] = \mathcal{O}(X^{2\sigma})\sum_k A_k^2$ *where* $A_k = 1/(\sigma^2 + \gamma_k^2)$*.*

(ii) The ratio $\mathrm{Var}_X[T_{12}]/T_{11}^2 = \mathcal{O}(X^{2\sigma-2}) \to 0$ *for all* $\sigma < 1$*. The tensor is asymptotically diagonal.*

(iii) Among all values of σ*,* $\sigma = 1/2$ *gives the* fastest *diagonalisation rate* ($\mathcal{O}(X^{-1})$)*.*

(iv) An off-line zero at $\sigma_0 > 1/2$ *contributes a term* $\mathcal{O}(X^{2\sigma_0})$ *that dominates the on-line variance* $\mathcal{O}(X)$*.*

Numerical verification (100 Odlyzko zeros, $X = 100$ to 10^5):

X	Var(σ=0.5)	Var(σ=0.6)	Ratio	$X^{0.2}$ (pred)
10^2	5.45	14.9	2.73	2.72
10^3	72.5	314	4.33	4.32
10^4	753	5,210	6.92	6.84
10^5	5,170	56,500	10.93	10.84

The ratio Var(0.6)/Var(0.5) matches the predicted $X^{2(0.6-0.5)} = X^{0.2}$ to better than 2% across three decades (Figure 2). (Script: `riemann_scaled_final.py`.)

4.3 The harmonic oscillator structure

Remark 4.2 (Stripped-mode ODE). Define the mode function $\varphi_k(t) = e^{\sigma t}[\sigma\cos(\gamma_k t) + \gamma_k \sin(\gamma_k t)]/(\sigma^2 + \gamma_k^2)$ and the envelope-stripped mode $y_k(t) = e^{-\sigma t}\varphi_k(t)$. Then for **all** $\sigma \in (0, 1)$:

$$y_k''(t) + \gamma_k^2\,y_k(t) = 0. \tag{15}$$

This is *definitionally true*: y_k is a linear combination of $\cos(\gamma_k t)$ and $\sin(\gamma_k t)$, so it automatically satisfies the SHO equation. The computation below merely confirms the algebra.

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1 Statement

Let $T_{B\text{–}L} = \mathrm{diag}(1/3, 1/3, 1/3, -1) \in \mathfrak{sl}(4, \mathbb{R})$ be the $B - L$ generator of the minimal Pati–Salam embedding, normalized so that the trace of the generator over the fundamental representation vanishes. The dimensionless quantities that arise mechanically from the bracket structure on $\mathfrak{sl}(4, \mathbb{R})$ with this generator are enumerated in Table 1.

Observation. Every entry in Table 1 lies in the multiplicative subgroup $\langle 2, 3 \rangle \subset \mathbb{Q}^\times$ generated by the primes 2 and 3. Equivalently, each native ratio is of the form $2^a \cdot 3^b$ for some $a, b \in \mathbb{Z}$.

This is mechanical. The 1/3 charge arises from $1/N_{\text{color}}$ with $N_{\text{color}} = 3$. The 4/3 gap is $1 - (-1/3) = 4/3 = 2^2 \cdot 3^{-1}$. The 16/9 saturation coefficient is the square of the gap. The multiplicities 3 and 9 are dimensions of color and color-squared structures. No prime greater than 3 enters the construction.

Remark (Connection to Pythagorean tuning). Pythagorean tuning is the system in which musical intervals are defined as the ratios in $\langle 2, 3 \rangle$, with 2 representing octave equivalence and 3 the lowest nontrivial harmonic [1]. Reduced into one octave by powers of two, the six native PS ratios of

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Figure 2: *Variance ratio* $\mathrm{Var}_X[T_{12}](\sigma{=}0.6)/\mathrm{Var}_X[T_{12}](\sigma{=}0.5)$ *computed with 50 Odlyzko zeros over three decades of* X*. The data (circles) match the predicted* $X^{0.2}$ *scaling (dashed line) to better than 2%. An off-line zero would produce a steeper power law.*

Verification. With $y_k = [\sigma\cos(\gamma_k t) + \gamma_k \sin(\gamma_k t)]/(\sigma^2 + \gamma_k^2)$:

$$y_k' = [-\sigma\gamma_k \sin(\gamma_k t) + \gamma_k^2 \cos(\gamma_k t)]/(\sigma^2 + \gamma_k^2),$$
$$y_k'' = [-\sigma\gamma_k^2 \cos(\gamma_k t) - \gamma_k^3 \sin(\gamma_k t)]/(\sigma^2 + \gamma_k^2) = -\gamma_k^2\, y_k.$$

Numerically confirmed: $|y_k'' + \gamma_k^2 y_k| < 9 \times 10^{-14}$ across 100 zeros, 10 values of t, and 5 values of $\sigma \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ (1,000 evaluations; `riemann_scaled_final.py`). □

The **non-trivial content** is not the individual-mode ODE (which is tautological) but the following:

Proposition 4.3 (Stationarity characterises the critical line)**.** *If all non-trivial zeros satisfy* $\mathrm{Re}(\rho) = 1/2$*, then every mode* φ_k *shares the envelope* $e^{t/2} = \sqrt{x}$*, so the relative amplitudes* A_k/A_j *are constant and the superposition* $F_N(x) = \sum_k A_k\varphi_k(x)$ *is stationary. If any zero has* $\mathrm{Re}(\rho) = \sigma_0 \neq 1/2$*, its envelope* x^{σ_0} *differs, and the superposition becomes non-stationary (the relative contribution of that mode grows or decays with* x*).*

Remark 4.4 (Self-correction from scaling)*.* An earlier version of this analysis stated that the harmonic-oscillator property held only at $\sigma = 1/2$. The scaled computation with real Odlyzko data revealed that $y_k'' + \gamma_k^2 y_k = 0$ holds for *all* σ (it must, by construction). The σ-dependence is entirely in the envelope $e^{\sigma t}$. The Riemann Hypothesis is therefore the statement: *the superposition of all zero modes is stationary* — every mode grows at the same rate.

4.4 Geometric vector fields and loop folding

The tensor (13) defines a vector field via the "curl" of the shear: $\omega_k = dT_{12}^{(k)}/dt$ where $t = \ln x$.

Proposition 4.5 (Pure rotation at $\sigma = 1/2$)**.** *At* $\sigma = 1/2$*, the curl of the shear for mode* k *reduces to:*

$$\omega_k\big|_{\sigma=1/2} = -2\sqrt{x}\,\cos(\gamma_k \ln x). \tag{16}$$

This is a pure cosine (zero radial component, pure rotation in the (t, γ) *plane). At* $\sigma \neq 1/2$*, a* sin *component appears with coefficient* $\gamma_k(\sigma - 1/2)$*, producing spiral flow (rotation + radial drift).*

Proof. Differentiate (14) with $\sigma = 1/2$: the sin term's coefficient is $\gamma(1/2 - 1/2) = 0$, leaving only the cos term with amplitude $2e^{t/2} = 2\sqrt{x}$. Verified symbolically: `riemann_tensor.py`. □

The topology of the flow:

- At $\sigma = 1/2$: purely rotational — closed loops (pinwheel). Fold lines (where the flow reverses) occur at $\gamma_k t = (n + 1/2)\pi$.
- At $\sigma > 1/2$: rotation + outward radial drift (opening spirals).
- At $\sigma < 1/2$: rotation + inward radial drift (closing spirals).

The critical line $\sigma = 1/2$ is the *unique center manifold* of this family of vector fields: the only value of σ where the flow is purely rotational with no radial component (Figure 3).

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Figure 3: *Tensor flow in the oscillatory plane for the first Riemann zero $\gamma_1 = 14.13$. Left: at $\sigma = 1/2$ the trajectory is a closed loop (pure rotation, no radial drift). Right: at $\sigma = 0.7$ the trajectory spirals outward (rotation plus radial growth from the $e^{(\sigma-1/2)t}$ envelope). The critical line is the unique value of σ where the flow is topologically closed.*

4.5 Wronskian non-vanishing at scale

The Wronskian $W[\varphi_k, \varphi_j] = \varphi_k \varphi_j' - \varphi_k' \varphi_j$ was evaluated at $\sigma = 1/2$, $t = 1$ for all $\binom{50}{2} = 1{,}225$ pairs of the first 50 Riemann zeros.

Result: $W \neq 0$ for all 1,225 pairs. Range: $|W| \in [8.6 \times 10^{-5},\ 0.19]$. Antisymmetry verified: $W[\varphi_k, \varphi_j] = -W[\varphi_j, \varphi_k]$ to machine precision. The Wronskian contains difference frequencies $(\gamma_k - \gamma_j)$ and sum frequencies $(\gamma_k + \gamma_j)$ — the combination tones of the arithmetic instrument. The incommensurability of the γ_k (from Montgomery–Odlyzko GUE statistics [4, 10]) ensures the Wronskian never vanishes.

Jacobi identity. The Wronskian bracket $[\varphi_i, \varphi_j] = W[\varphi_i, \varphi_j]$ satisfies the Jacobi identity: $[\varphi_i, [\varphi_j, \varphi_k]] + [\varphi_j, [\varphi_k, \varphi_i]] + [\varphi_k, [\varphi_i, \varphi_j]] = 0$, verified numerically to relative precision 3×10^{-10} across 10 triples of the first 5 zeros at 4 values of t (limited by numerical differentiation; `riemann_tensor.py`). Combined with antisymmetry and non-degeneracy, this confirms that the Wronskian defines a genuine infinite-dimensional Lie algebra on the space of Riemann zero modes (Figure 4).

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Figure 4: *Wronskian matrix $W[\varphi_k, \varphi_j]$ evaluated at $t = 1$, $\sigma = 1/2$ for the first 20 Riemann zeros. The antisymmetric structure ($W_{kj} = -W_{jk}$) is visible; no entry is zero. The checkerboard pattern reflects the alternation of difference-frequency signs across the zero spectrum.*

5 Finite-sample stability

The results of Sections 2–2.6 were originally verified using the first 50 Odlyzko zeros. We report here an extension to $N = 200$ exact zeros (computed via the Riemann–Siegel formula to 15-digit precision), which removes small-sample doubt from the empirical claims.

Wronskian non-vanishing. The non-vanishing Wronskian condition (Proposition 2.2) was verified for all $\binom{200}{2} = 19{,}900$ distinct pairs of zeros, with no exceptions observed. This is a $16\times$ extension of the original $\binom{50}{2} = 1{,}225$-pair test. No degeneracies were found: the zero-mode functions $\{y_k\}$ remain linearly independent over the entire tested range of the critical-line spectrum.

Cross-correlation decay. The cross-correlation bound $|C_N| = \sup_x \left| \frac{2}{N(N-1)} \sum_{j<k} \cos\big((\gamma_j - \gamma_k)x\big) \right|$ was measured at $|C_{200}| = 0.044$, well below the $|C_N| < 0.29$ threshold required for the converse (Theorem 3.2). The bound improves with increasing N, consistent with cancellation of off-diagonal contributions in the pair-correlation sum.

Arithmetic consequence: prime gap transition statistics. The cross-correlation bound $|C_{200}| = 0.044$ admits a quantitative bridge to an empirical observable in prime distribution data. In [2] the second-order Dynamical Dynamical Dynamical Dynamical Dynamical Transition Operators over Prime Gap Ensembless over Prime Gap Ensembless over Prime Gap Ensembless over Prime Gap Ensembless over Prime Gap Ensembles on the multiplicative residue group $(\mathbb{Z}/m\mathbb{Z})^*$ is constructed from 5×10^7 primes up to 10^9. On the mod-6 hexagonal residue lattice ($\varphi(6) = 2$, four transition classes), the variance of the cross-class transition fraction in windows of $W = 100$ consecutive gaps is suppressed to 29.2% of the value expected under independent sampling. This empirical floor satisfies $R = 1/(1 + (N_{\text{eff}} - 1) \cdot |C_N|) = 1/(1 + 55 \times 0.044) = 0.292$ to three significant figures, with $N_{\text{eff}} = 56$ extracted from the empirical autocorrelation structure of the cross-class indicator. The Wronskian non-vanishing established here is the spectral-side input; the arithmetic variance floor is the output. Details and the operator-theoretic context, including the empirical Kernel Law $\dim\ker(P_m) = \varphi(m)$ and three documented falsified conjectures, appear in [2].

Stationarity at scale ($N = 2{,}000{,}000$). Extension to the full $N = 2{,}001{,}052$ zeros of Odlyzko's `zeros6` dataset confirms the stationarity prediction of Theorem 2.1 directly across six decades of N. We sample $F_N(x)$ at $M = 10{,}000$ uniformly spaced points on $[0, X_{\max}]$ for $X_{\max} \in \{10, 10^2, 10^3, 10^4, 10^5\}$ and extract the log-log scaling exponent α_N from $\log(\mathrm{var}(F_N)) \approx \alpha_N \log(X_{\max}) + \text{const}$ by ordinary least squares. The unperturbed exponent is tightly bounded near zero across six decades of N:

$$\begin{aligned} \alpha(50) &= -0.000200 & \alpha(10{,}000) &= +0.001045 \\ \alpha(100) &= -0.000186 & \alpha(100{,}000) &= +0.001247 \\ \alpha(200) &= +0.000875 & \alpha(2{,}000{,}000) &= +0.001259 \end{aligned}$$

giving $\alpha \approx 0$ with deviation bounded by 1.3×10^{-3} across the full range tested. The variance of F_N is empirically bounded as the window grows, in direct agreement with the forward direction of Theorem 2.1. The flatness of α_N across four additional orders of magnitude in N beyond previous tests is evidence that the diagnostic is measuring a structural property of the spectrum rather than a finite-N numerical artefact.

Off-line injection scaling. To verify that the variance observable is sensitive to off-line contributions and not merely insensitive to everything, we introduce a single synthetic zero at height $\gamma_{\text{fake}} = 1000$ with an amplitude that breaks critical-line stationarity for $\sigma \neq 1/2$:

$$F_N^{\text{perturbed}}(x) = F_N(x) + \frac{x^{\sigma-1/2}}{\frac{1}{4}+\gamma_{\text{fake}}^2}\,\varphi_{\text{fake}}(x). \tag{17}$$

The factor $x^{\sigma-1/2}$ vanishes at $\sigma = 1/2$ (consistent with stationarity on the critical line) and grows or decays as a power law otherwise. The predicted scaling exponent of the perturbed variance is $\alpha = 2\sigma - 1$.

σ_{inject}	Predicted $2\sigma - 1$	Measured α
0.40	-0.200	-0.197
0.45	-0.100	-0.099
0.55	$+0.100$	$+0.099$
0.60	$+0.200$	$+0.199$

Table 2: *Empirical scaling exponent under off-line injection at $N = 2{,}000{,}000$.*

All four off-line cases recover the predicted exponent to within 1.5% (best case 0.5% at $\sigma = 0.60$). Combined with the unperturbed $\alpha_N \approx 0$ across six decades of N, this empirically validates the variance observable as a sensitive diagnostic for off-line spectral contributions over the σ-range tested. The convergence of measured exponents toward the exact predicted values as N grows (from $\sim 2\%$ at $N = 100{,}000$ to $\sim 1\%$ at $N = 2{,}000{,}000$) is consistent with $\alpha = 2\sigma - 1$ being an asymptotic identity rather than a finite-N coincidence. The scaling result is not a proof of the converse of Theorem 2.1, but it establishes that the diagnostic behaves at $N = 2{,}000{,}000$ exactly as the theorem predicts.

Cross-correlation at scale. The cross-correlation bound at $N = 2{,}001{,}052$:

$$|C_{2{,}001{,}052}| = 7.00 \times 10^{-6},$$

compared to $|C_{200}| = 0.044$. Under independent random spacings, the decay would follow $|C_N| \sim N^{-1/2}$, predicting $|C_{2{,}001{,}052}| \approx 4.4 \times 10^{-4}$ from the $N = 200$ baseline. The observed bound is approximately $63\times$ smaller, fitting a power law

$$|C_N| \sim N^{-0.95}.$$

The decay exponent has sharpened from $N^{-0.83}$ at $N = 10^5$ toward the deterministic-cancellation bound N^{-1} as N grows, indicating that the rigidity mechanism strengthens rather than saturates at scale. The accelerated decay is qualitatively consistent with the Gaussian Unitary Ensemble (GUE) rigidity of Riemann zero spacings: the correlated spacing structure produces stronger destructive interference in the off-diagonal pair sum than independent oscillators would, and the strength of this interference is itself a function of N that approaches its theoretical maximum at the scale of Odlyzko's full reference dataset.

6 Tartini-Tone Characterization of the Spectrum

The cross-correlation result $|C_N| \sim N^{-0.95}$ in Section 5 admits a clean reformulation in spectral-musical language. The function $C_N(x)$ is, up to normalisation, the empirical characteristic function of the difference-frequency distribution $\{\gamma_k - \gamma_j : k > j\}$ — what acoustics calls the *Tartini difference-tone spectrum* of the zero set. Two distinct frequencies γ_k and γ_j processed through any nonlinear element generate combination tones at $\gamma_k \pm \gamma_j$, $2\gamma_k - \gamma_j$, and so forth. The empirical pair-correlation, the explicit-formula peak structure, and the cross-correlation bound are three observables of the same Tartini-tone density viewed through different test functions.

We extended this characterisation in three directions empirically, testing the framework's structural claim that the prime sequence (Form) and zero sequence (Function) are Fourier-dual asymmetric codependent partners.

6.1 Order-2: Montgomery's pair correlation

The order-2 normalised pair density was computed at $N = 5{,}000$ zeros (12,497,500 pairs). The empirical density reproduces Montgomery's GUE pair-correlation conjecture

$$R_2(\alpha) = 1 - \left(\frac{\sin \pi\alpha}{\pi\alpha}\right)^2 \tag{18}$$

to within statistical noise. Empirical R_2 at $\alpha \approx 1$ yields 0.97 (theory 1.00); at $\alpha \approx 3$ yields 0.98 (theory 1.00). The repulsion at $\alpha \to 0$ rises as $(\pi\alpha)^2/3$ as predicted. This is the Tartini difference-tone density of the Riemann zero spectrum, and confirms GUE-class 2-level rigidity at our scale.

6.2 Order-3: Rudnick–Sarnak 3-level correlation

The order-3 correlation surface $R_3(\alpha, \beta)$ was computed at $N = 50{,}000$ zeros with 606,189 normalised consecutive-difference triples on a 40×40 grid covering $\alpha, \beta \in [0, 4]$. The empirical surface is compared to the Rudnick–Sarnak GUE prediction

$$R_3(\alpha, \beta) = 1 - s(\alpha)^2 - s(\beta)^2 - s(\alpha + \beta)^2 + 2\, s(\alpha)\, s(\beta)\, s(\alpha + \beta), \tag{19}$$

where $s(x) = \sin(\pi x)/(\pi x)$ [13]. The empirical match is

Metric	Value
RMSE vs theory	0.051
Max |residual|	0.182
Pearson correlation	0.990

Sample cross-section comparisons confirm the agreement:

(α, β)	theory	empirical	ratio
$(0.5, 0.5)$	0.189	0.154	0.81
$(1.0, 1.0)$	1.000	1.044	1.04
$(1.5, 0.5)$	0.550	0.538	0.98
$(2.0, 2.0)$	1.000	0.880	0.88
$(3.0, 3.0)$	1.000	0.948	0.95

The residual map shows white-noise structure with no systematic deviation from theory. This confirms GUE-class 3-level rigidity.

6.3 Fourier-dual: explicit-formula peak structure

The deepest connection between primes and zeros is the Riemann–Weil explicit formula [14], which expresses sums over zeros as Fourier-transform-dual to sums over prime powers with von Mangoldt weights:

$$\sum_k h(\gamma_k) \approx (\text{smooth}) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \hat{h}(\log n). \tag{20}$$

Taking the test function $h(t) = \cos(\omega t)$ and defining

$$F_N(\omega) := \frac{1}{N} \sum_{k=1}^{N} \cos(\omega \gamma_k), \tag{21}$$

the explicit formula predicts that $F_N(\omega)$ develops peaks at $\omega = \log(p^k)$ for prime powers p^k, with relative heights $-\Lambda(p^k)/\sqrt{p^k}$ in the asymptotic limit.

Empirically, at $N = 10^5$ zeros over $\omega \in (0, 8]$ with 2000 grid points, the top 20 peaks of $|F_N|$ all align with prime-power positions to within 1.8×10^{-4}, with mean distance 5×10^{-5}. The random-baseline mean distance (uniformly distributed test points) is 4.3×10^{-2}. The ratio is 0.0012 (peaks are 833 times closer), with KS test $p < 10^{-4}$. Selected top peaks:

rank	ω_{peak}	$\lvert F_N \rvert$	nearest prime power
1	6.13340	0.0335	461 (prime)
2	5.93755	0.0312	379 (prime)
3	6.71297	0.0251	823 (prime)
10	**4.15888**	**0.0103**	$\mathbf{2^6 = 64}$ (prime power)
12	4.11092	0.0062	61 (prime)
15	3.36748	0.0049	29 (prime)

The presence of $2^6 = 64$ at rank 10 is critical: it confirms that the peaks are at *prime powers*, not just primes, exactly as the von Mangoldt weighting in (20) predicts. This is the Riemann–Weil explicit formula made spectrally visible.

6.4 Synthesis

The three observables R_2, R_3, and the Fourier-dual peak alignment form a complete Tartini-tone fingerprint of the spectrum:

- **Internal rigidity**: R_2 and R_3 characterise the zero set's GUE-class spacing statistics.
- **External coupling**: $F_N(\omega)$ peak structure characterises the Fourier-dual coupling to the primes through the explicit formula.

Crucially, no direct resonance between the zero spectrum and the prime spectrum is observed at first order: the lowest zero $\gamma_1 = 14.135$ sits above the maximum first-order prime frequency $\max_p 2\pi/\ln p = 9.06$, and tests for direct alignment between γ_k and combinations of prime log-frequencies $\sum_i \ln p_i$ at orders 2 and 3 produce p-values consistent with the null hypothesis [17]. The coupling is realised entirely through the explicit-formula transformation, not through frequency matching.

This empirical pattern is the ACS framework's structural claim made spectrally explicit: the prime sequence and zero sequence are *Fourier-dual asymmetric codependent partners*, exchanging information through transformation rather than through resonant alignment.

7 L-Function Generalisation

The Tartini-tone characterisation extends from the Riemann zeta function to other L-functions in the Selberg class. We tested three non-trivial cases.

7.1 Dirichlet L-function $L(s, \chi_4)$

The Dirichlet L-function with the unique non-trivial character mod 4, $\chi_4(n) = 1$ for $n \equiv 1 \pmod 4$, $\chi_4(n) = -1$ for $n \equiv 3 \pmod 4$, $\chi_4(2) = 0$, has the explicit-formula generalisation

$$\sum_k h(\gamma_k^L) \approx (\text{smooth}) - \sum_{n\geq 2} \frac{\chi_4(n)\Lambda(n)}{\sqrt{n}} \hat{h}(\log n). \tag{22}$$

The key difference: the character χ_4 enters as a sign assignment on the prime-power contributions.

We computed 122 zeros of $L(s, \chi_4)$ on the critical line by detecting sign changes of the completed function $\Lambda(s, \chi_4) = (4/\pi)^{s/2}\Gamma((s+1)/2)L(s, \chi_4)$ (real on the critical line for real characters), then refining via bisection to 10^{-10} precision. Range: $[\gamma_1^L, \gamma_{122}^L] = [6.02, 198.79]$.

Defining $F_L(\omega) = (1/N)\sum_k \cos(\omega\gamma_k^L)$, the explicit formula (22) predicts:

- Positive F_L peaks at $\omega = \log(p^k)$ where $\chi_4(p^k) = -1$ (due to the leading minus sign in the explicit formula).
- Negative F_L peaks at $\omega = \log(p^k)$ where $\chi_4(p^k) = +1$.

Empirically, the top 19 peaks of $|F_L|$ for $\omega \in [0.5, 5.0]$ yield $19/19 = 100\%$ sign accuracy (binomial $p = 2 \times 10^{-5}$), with alignment distances of 0.001 to 0.004 between peak positions and prime-power log-frequencies. The Dirichlet character χ_4 is *recovered* from the L-function's zeros through its Fourier transform, including for composite prime powers: $3^3 = 27$ (where $\chi_4(27) = (-1)^3 = -1$) and $5^2 = 25$ (where $\chi_4(25) = (+1)^2 = +1$).

7.2 Cross-correlation: Dirichlet's theorem via Fourier duality

Combining $F_\zeta(\omega)$ from the Riemann zeros and $F_L(\omega)$ from $L(s, \chi_4)$ produces an arithmetic decomposition. At $\omega = \log(p^k)$:

$$F_\zeta(\omega) \propto -\frac{\Lambda(p^k)}{\sqrt{p^k}}, \tag{23}$$

$$F_L(\omega) \propto -\frac{\chi_4(p^k)\Lambda(p^k)}{\sqrt{p^k}}. \tag{24}$$

Hence:

$$\frac{F_\zeta - F_L}{2} \propto -\frac{1-\chi_4(p^k)}{2} \cdot \frac{\Lambda(p^k)}{\sqrt{p^k}} = \begin{cases} 0 & \chi_4 = +1 \text{ (primes} \equiv 1 \text{ mod } 4) \\ -\Lambda(p^k)/\sqrt{p^k} & \chi_4 = -1 \text{ (primes} \equiv 3 \text{ mod } 4) \end{cases} \tag{25}$$

$$\frac{F_\zeta + F_L}{2} \propto -\frac{1+\chi_4(p^k)}{2} \cdot \frac{\Lambda(p^k)}{\sqrt{p^k}} = \begin{cases} -\Lambda(p^k)/\sqrt{p^k} & \chi_4 = +1 \\ 0 & \chi_4 = -1 \end{cases} \tag{26}$$

This is the standard Dirichlet decomposition used in the proof of Dirichlet's theorem on primes in arithmetic progressions (1837) [6], viewed through the Fourier-dual lens.

Empirically: the top 15 peaks of $(F_\zeta - F_L)/2$ all correspond to primes $\equiv 3 \bmod 4$ $(3, 7, 11, 19, 23, 31, 43, 47, 59, 67, 71, 79, 83, 107, 127)$. The top 15 peaks of $(F_\zeta + F_L)/2$ all correspond to primes $\equiv 1 \bmod 4$ $(5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, 101, 109, 113, 137)$. Both decompositions yield $15/15 = 100\%$ accuracy.

Powers of 2 (where $\chi_4(2) = 0$) appear in both $(F_\zeta - F_L)/2$ and $(F_\zeta + F_L)/2$ with equal amplitude: at $\omega = \log 2$, the ratio is 0.95; at $\omega = \log 4$, the ratio is 0.99. This is the expected behaviour for the ramified prime in the arithmetic of $\mathbb{Q}(i)$, foreshadowing the Dedekind structure.

7.3 Dedekind zeta of $\mathbb{Q}(i)$

The Dedekind zeta function of the field of Gaussian integers factorises as [15]

$$\zeta_K(s) = \zeta(s) \cdot L(s, \chi_4) \quad \text{for } K = \mathbb{Q}(i). \tag{27}$$

Equivalently, the zeros of ζ_K are the union of the zeros of ζ and $L(s, \chi_4)$. The prime-ideal structure of $K = \mathbb{Q}(i)$ is:

- The prime 2 ramifies: $(2) = (1 + i)^2$. Norm of $(1 + i)$ is 2.
- Primes $p \equiv 1 \pmod 4$ split: $(p) = \mathfrak{p}\bar{\mathfrak{p}}$, each prime ideal $\mathfrak{p}$ has norm p.
- Primes $p \equiv 3 \pmod 4$ remain inert: (p) is a single prime ideal of norm p^2.

The Fourier transform $F_K(\omega) = F_\zeta(\omega) + F_L(\omega)$ of the Dedekind zero set should therefore have peaks at:

- $\omega = \log 2$ (ramified, single ideal of norm 2)
- $\omega = \log p$ for $p \equiv 1 \pmod 4$ (split, with doubled weight from two ideals)
- $\omega = \log p^2$ for $p \equiv 3 \pmod 4$ (inert, ideal has norm p^2, not p)

And crucially, *no peak* at $\omega = \log p$ for $p \equiv 3 \pmod 4$, since these primes have no ideal of norm p.

Empirically, 19/20 of the top peaks of $F_K(\omega)$ align with predicted ideal-norm positions within 0.05. The amplitude pattern is:

Position class	Mean $\lvert F_K \rvert$	Predicted
$\log p$, $p \equiv 1 \pmod 4$ (split)	0.323	high
$\log p$, $p \equiv 3 \pmod 4$ (inert)	0.057	near zero
$\log p^2$, $p \equiv 3 \pmod 4$ (inert)	0.144	restored

The ratio of split to inert (at $\log p$) is $5.7\times$. The algebraic signature of $\mathbb{Z}[i]$ — the splitting pattern of primes — is recovered empirically from the zero Fourier transform.

7.4 Status of the generalisation

The framework's claim of Fourier-dual asymmetric codependence extends from $\zeta(s)$ to:

1. Dirichlet L-functions with non-trivial real character (χ_4 recovered as sign assignment, 100% accuracy on 19 peaks).
2. Cross-correlations of L-functions (decomposing primes by residue class mod 4, 100% accuracy on 30 peaks).
3. Dedekind zeta of imaginary quadratic field ($\mathbb{Q}(i)$'s prime-ideal structure recovered, $5.7\times$ split/inert amplitude ratio).

These are not new mathematical theorems; they re-derive classical results (Dirichlet 1837, Weil 1952, Rudnick–Sarnak 1996) in the framework's specific Fourier-dual framing. What is new is the unified operational description: the zero spectrum of any L-function in the Selberg class is the Fourier dual of its prime-side arithmetic structure, with the character, splitting pattern, or arithmetic invariant encoded as sign/amplitude assignments on Fourier-transform peaks.

The natural extensions, all computationally tractable using the same methodology, are:

- Other Dirichlet characters χ_q for $q \in \{3, 5, 7, 8, \ldots\}$.
- Imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$ for additional discriminants d.
- Real quadratic fields.
- Cyclotomic fields $\mathbb{Q}(\zeta_n)$, factorising over the full character group mod n.
- Higher-degree L-functions (modular, automorphic).

8 Hilbert–Pólya Constraint Specification

The Hilbert–Pólya conjecture posits the existence of a self-adjoint operator H on some Hilbert space whose spectrum is exactly the set of imaginary parts of non-trivial Riemann zeros, $\{\gamma_k\}$. If such H exists, the Riemann Hypothesis follows immediately (real eigenvalues of self-adjoint operators).

The empirical fingerprint of the zeros, characterised in Sections 5–7, provides a constraint specification for any candidate operator. We make this specification explicit.

8.1 The constraints

A candidate spectrum $\{\lambda_k\}$ (claimed to coincide with $\{\gamma_k\}$) must satisfy:

ID	Constraint
C1	Density: $N(T) \sim (T/2\pi)\log(T/2\pi) - T/2\pi$
C2	Variance stationarity: $\alpha(N) \to 0$ as $N \to \infty$
C3	Pair correlation: $R_2(\alpha) = 1 - (\sin \pi\alpha/\pi\alpha)^2$
C4	3-level: R_3 matches Rudnick–Sarnak GUE form
C5	Fourier-dual: $F_N(\omega)$ peaks at $\omega = \log(p^k)$
C6	Sign: peaks at $\log(p^k)$ are negative
C7	Weights: peak heights $\propto \Lambda(p^k)/\sqrt{p^k}$
C8	L-function transfer: signs follow $\chi(p^k)$ for arithmetic operators
C9	Selberg orthogonality: distinct L-function operators give orthogonal F

Operationally: H satisfies the constraint specification if $\mathrm{Tr}(\cos(\omega H))$ reproduces the prime explicit formula *and* the spectral statistics follow GUE. The two halves are *independent* constraints; both must hold.

8.2 GUE alone is necessary but not sufficient

A common misconception is that finding an operator with GUE-distributed eigenvalues at the correct density suffices. It does not.

Generating 1000 eigenvalues of a random 2000×2000 GUE matrix (complex Hermitian with i.i.d. Gaussian entries) and linearly rescaling to match the Riemann range $[14.13, 1419.42]$, we obtain a spectrum with empirical pair correlation indistinguishable from the actual Riemann zeros:

Spectrum	R_2 RMSE	Fourier-dual alignment ratio
Riemann zeros (truth)	0.597	0.012
GUE matrix eigenvalues	0.600	0.495

The pair correlations are statistically identical. The Fourier-dual alignment differs by a factor of 40: Riemann zero peaks land $\approx 100\times$ closer to prime-power positions than random test points, while GUE matrix eigenvalues are essentially random with respect to $\{\log(p^k)\}$.

This rules out a vast class of candidate operators. Any self-adjoint matrix in the GUE ensemble gives GUE-distributed eigenvalues; none of them produce the Fourier-dual prime structure. The Hilbert–Pólya operator must be *specific*: its trace formula must equal the prime explicit formula.

8.3 The Selberg trace formula reading

The Fourier-dual constraint (C5–C7) is structurally a Selberg-type trace formula condition. For a hyperbolic surface $\Gamma\backslash\mathbb{H}$, the Selberg trace formula [16] relates Laplacian eigenvalues (spectral side) to lengths of closed geodesics (geometric side):

$$\sum_n h(r_n) = (\text{identity contribution}) + \sum_{\{\gamma\}} \frac{\ell(\gamma_0)\,\hat{h}(\ell(\gamma))}{2\sinh(\ell(\gamma)/2)}. \tag{28}$$

For the Hilbert–Pólya problem, we have empirical access to both sides:

- Spectral side: the Riemann zeros $\{\gamma_k\}$.
- Geometric side: $\log(p^k)$ with von Mangoldt weights $\Lambda(p^k)/\sqrt{p^k}$.

What is missing is the actual geometric space whose closed orbits have lengths at log(prime power). Several proposals exist (Berry–Keating $xp + px$ [7]; Connes' adèle space [8]; Wu–Sprung Schrödinger candidates) but none have rigorously produced the full constraint set.

8.4 The framework's structural hints

The ACS framework's distinctive contribution to this problem is the suggestion that H should have an internal Form–Function pairing, rather than be a single self-adjoint operator on a single space. Three operator-theoretic translations are consistent with the empirical data but not yet rigorously constructed:

Direct sum with asymmetric coupling. $H = H_{\text{Form}} \oplus H_{\text{Function}} + H_{\text{coupling}}$ on $\mathcal{H}_{\text{Form}} \oplus \mathcal{H}_{\text{Function}}$, where H_{coupling} is the asymmetric off-diagonal piece realising the Fourier-dual coupling.

Tensor product with Lie algebra coupling. H acts on $\mathcal{H}_{\text{Form}} \otimes \mathcal{H}_{\text{Function}}$ with the coupling living in the BCH expansion of an underlying Lie algebra (the ACS Palatini bracket framework [1]).

Graded operator with Koszul duality. H is a graded operator on a Koszul-dual pair of spaces (Form-graded and Function-graded), where the explicit formula corresponds to the duality between the two gradings.

These hints are conjectural structural specifications. They are not operator constructions. Their purpose is to constrain the search: any candidate H should exhibit Form–Function asymmetric codependent structure, not just self-adjointness.

8.5 Executable test suite

The constraint specification is reduced to executable code. For any candidate spectrum $\{\lambda_k\}$ supplied as a NumPy array, the test suite [18] computes:

- α_N for variance stationarity (C2)
- R_2 RMSE versus Montgomery (C3)
- R_3 RMSE versus Rudnick–Sarnak (C4)
- Mean Fourier-dual peak distance to $\{\log(p^k)\}$ (C5)
- Sign accuracy at $\omega = \log p$ for small primes (C6)

Each test returns pass/fail with calibrated thresholds derived from the actual Riemann zero data.

Scorecard on test spectra. Four candidate spectra:

Spectrum	C1	C3 (R_2)	C5 (Fourier)	C6 (sign)
Riemann zeros (truth)	PASS	PASS	PASS	PASS
GUE matrix eigenvalues	PASS	PASS	FAIL	FAIL
Random/Poisson with matching density	PASS	FAIL	FAIL	FAIL
Equally spaced points	PASS	FAIL	FAIL	FAIL

Only the actual Riemann zeros pass all constraints. The Berry–Keating $xp + px$ heuristic and other published proposals have not yet been tested against the full constraint set; doing so is a tractable open direction.

8.6 Status

We have not constructed the Hilbert–Pólya operator. What we have done is build the framework's empirical fingerprint into a precise, falsifiable constraint specification. The target is no longer the loose "find an operator with eigenvalues γ_k" but the sharp **find a self-adjoint H such that $\mathrm{Tr}(\cos(\omega H))$ reproduces the prime explicit formula and spectral statistics follow GUE.**

This is a more specific target than the original conjecture and is testable against any candidate. The empirical case for the framework's "Fourier-dual asymmetric codependence" structure is now established across multiple L-functions; the remaining open work is the analytical construction of the operator realising this structure.

9 Open Problems and Status

We summarise the epistemic status of all results.

Proved. RH $\Rightarrow$ stationarity of F_N (Theorem 2.1, via the AM–GM envelope bound). The Wronskian is a Lie bracket in function space (Proposition 2.2). The stripped mode satisfies $y_k'' + \gamma_k^2 y_k = 0$ for all σ (Theorem 4.2). The curl of the shear is a pure cosine at $\sigma = 1/2$ (Proposition 4.5). Renormalised stability under RH (Theorem 2.4, the logarithmic-coordinate restatement of von Koch's bound [11]).

Confirmed (real data, 100 Odlyzko zeros). The converse (Theorem 3.2: stationarity $\Rightarrow$ RH) is rigorous for any fixed N; the extension to all N requires $\delta_N > 0$, verified to $N = 200$. Variance scaling matches $X^{2\sigma}$ to $< 2\%$ over three decades (Theorem 4.1). Wronskian non-vanishing on all 1,225 pairs of 50 zeros. The critical line is the unique center manifold of the tensor flow. Resolvent identity $\chi(\omega) = \mathrm{Tr}[(\omega - H)^{-1}]$ for the trivial diagonal H (verified at machine precision in `src/paper_b/explicit_formula_resolvent.py`). Berry–Keating leading semiclassical counting [7] matches the Riemann–von Mangoldt density to within ~ 1 zero through $T = 143$ (`src/paper_b/berry_keating_counting.py`).

Disproved (negative results, scope of model space). The Wronskian on scalar zero-mode functions *fails the Leibniz axiom* by $W[fg, h] - (f\,W[g, h] + g\,W[f, h]) = -fgh'$ (Remark 2.3, verified symbolically). Therefore the Wronskian is not a Poisson bracket on a function algebra; plasma-Hamiltonian readings of the Wronskian-as-bracket structure do not transfer. The phenomenological correspondences with discrete plasma resonances (dispersion peaks at $\omega \approx \gamma_k$, beat-frequency Wronskian zero crossings, neutral-stability framings) survive as analogy only; the symplectic-foundation reading does not.

Self-corrections. The harmonic-oscillator property $y'' + \gamma^2 y = 0$ was initially reported as holding "only at $\sigma = 1/2$"; the scaled computation with real data showed it holds for *all* σ (Remark 4.4). The corrected statement: the critical line is distinguished by *stationarity of the superposition*, not by the individual mode ODE. An earlier draft proposed a "plasma Hamiltonian" framing of the prime–zero system as the foundation of the spectral stationarity result; subsequent verification of the Wronskian's failure of Leibniz (Remark 2.3) showed this framing is not supported by the algebra. The result has been corrected to use the resolvent-trace formulation (Section 2.5) and the renormalised-stability formulation (Section 2.6), both of which are rigorous and standard.

Open directions.

Extension to L-functions. Section 7 extends the framework to non-zeta L-functions empirically: $L(s, \chi_4)$ recovers the Dirichlet character χ_4 as sign assignment on $F_L(\omega)$ peaks (100% accuracy on 19 peaks); the cross-correlation $(F_\zeta \pm F_L)/2$ decomposes primes by residue class mod 4 (100% on 30 peaks); and $\zeta_K(\mathbb{Q}(i))$ recovers the split/inert prime-ideal structure of $\mathbb{Z}[i]$ (5.7× amplitude ratio). What remains open is broader generalisation across the Selberg class: characters χ_q for $q \in \{3, 5, 7, 8, \ldots\}$; imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$; real quadratic fields; cyclotomic fields $\mathbb{Q}(\zeta_n)$; and higher-degree (modular, automorphic) L-functions. The same methodology applies; only the zero-finding step requires adaptation per case.

Unconditional converse. Removing the $\delta_N > 0$ condition in Theorem 3.2 would make the equivalence RH $\Leftrightarrow$ stationarity unconditional.

Higher-N verification. Section 5 verifies the stationarity prediction at $N = 2{,}000{,}000$ on Odlyzko's `zeros6` reference dataset of 2,001,052 zeros: $\alpha_N \approx 0$ for the unperturbed spectral function across six decades of N, and the injection scaling $\alpha = 2\sigma - 1$ is recovered to 1.5% accuracy across four off-line σ values. The diagnostic is empirically validated at the largest publicly available scale; what remains open is the unconditional converse (above).

Hilbert–Pólya operator construction. Find a natural self-adjoint H (Schrödinger or otherwise) whose spectrum is exactly $\{\gamma_k\}$, not inputted. This is the 100-year-old Hilbert–Pólya problem [7, 8]; resolution would upgrade the resolvent-trace framing of Section 2.5 from formal identity to derived theorem. Section 8 provides the framework's contribution: a precise constraint specification (C1–C9) with executable test suite, and the demonstration that GUE statistics alone are necessary but not sufficient (Riemann zeros and random Hermitian eigenvalues are indistinguishable on R_2 but differ by 40× on the Fourier-dual peak alignment). The target is sharpened to: *find H such that* $\mathrm{Tr}(\cos(\omega H))$ *reproduces the prime explicit formula and spectral statistics follow GUE.* Three structural hints from the ACS algebra (direct sum with asymmetric coupling, tensor product with Lie algebra coupling, graded operator with Koszul duality) are documented but await rigorous operator-theoretic realisation.

10 Discussion

The elastodynamic tensor analysis reveals a coherent geometric picture of the Riemann explicit formula, derived without assuming RH.

The critical line $\sigma = 1/2$ emerges as the *unique center manifold* of a well-defined dynamical system. Each Riemann zero γ_k is the natural frequency of a harmonic oscillator (Theorem 4.2): the envelope-stripped mode y_k satisfies $y_k'' + \gamma_k^2 y_k = 0$ independently of σ. The full mode $\varphi_k = e^{\sigma t} y_k$ grows with envelope x^σ. The Riemann Hypothesis is the statement that all modes share the same envelope ($x^{1/2}$), making the superposition $F_N = \sum_k A_k \varphi_k$ *stationary* — the relative amplitudes are constant in time.

The tensor field generated by the explicit formula has a purely rotational structure at $\sigma = 1/2$ (Proposition 4.5): the curl of the shear stress is a pure cosine with no radial drift. Away from the critical line, a radial component appears (proportional to $\sigma - 1/2$), converting the closed loops into spirals. The critical line is therefore the unique value of σ where the tensor flow is topologically closed — a pinwheel rather than a spiral.

The Wronskian of the zero modes is non-vanishing on all 1,225 tested pairs, establishing an infinite-dimensional Lie algebra on the space of modes. The combination frequencies $\gamma_k \pm \gamma_j$ appearing in the Wronskian are the "Tartini tones" of the arithmetic instrument: objective nonlinear mixing products generated by the coupled prime–zero system.

Relation to the Hilbert–Pólya programme. The idea that RH is equivalent to the self-adjointness of a specific operator dates to Hilbert and Pólya. Berry and Keating [7] proposed the quantisation of xp as a candidate; Connes [8] gave a trace-formula approach via noncommutative geometry. Our stationarity characterisation (Theorem 3.2) is in this tradition: it replaces "self-adjointness of an operator" with "stationarity of a superposition," but the underlying logic is the same — a spectral condition on the zeros equivalent to RH. The recent work of Baluyot and Pratt [9] on the Alternative Hypothesis (what if zeros are *not* on the critical line but instead spaced at half-integers?) provides the natural test case: under the AH, the variance diagnostic of Theorem 4.1 would exhibit a characteristic signature distinct from the RH prediction, offering a potential discriminant between the two hypotheses. The pair correlation results of Baluyot et al. [10] provide the rigorous background for the Wronskian analysis of Section 4.

The variance scaling (Theorem 4.1), confirmed with 100 Odlyzko zeros to $<2\%$ accuracy over three decades, provides a quantitative diagnostic: an off-line zero at $\sigma_0 > 1/2$ would produce a variance excess scaling as $X^{2\sigma_0 - 1}$ relative to the on-line contribution. This is in principle detectable by high-precision numerical computation of $\mathrm{Var}_X[T_{12}]$ at large X.

All conclusions follow from the tensor geometry and the real Riemann zero data. The critical line's special role is not assumed but emerges as a mathematical consequence of the structure: it is where the oscillators are in tune, the flow is closed, and the variance is minimised.

Reproducibility

All computational claims are backed by the scripts `riemann_tensor.py` and `riemann_scaled_final.py`, executable with Python 3.8+ and the standard libraries NumPy, SciPy, and SymPy. The

100 Riemann zeros are from the Odlyzko high-precision tables (hardcoded for exact reproducibility). See the accompanying `README_verification_suite.md` for a complete script inventory.

A Geometric Vision

The prime numbers are the rigid boundary of arithmetic — the instrument. The Riemann zeros are the resonant frequencies — the vibrations. The explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho - \ldots$ is the coupling equation that ties them together: primes (Form) and zeros (Function) in mutual constraint.

When all zeros sit on the critical line ($\sigma = 1/2$), every mode has the same amplitude envelope. The superposition is stationary — the instrument is perfectly in tune. An off-line zero would be a mistuned harmonic: its mode grows or decays, destroying the stationarity.

The prime–zero ACS is the arithmetic instance of a nesting that runs across all domains: vibrations (zeros) $\to$ momentum (bracket coupling) $\to$ gauge colour ($\mathfrak{su}(3)$ attractor) $\to$ confinement $\to$ mass $\to$ gravity.

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